

# Sierra Manriquez

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## Summary

Third-year Mechanical Engineering student with research, mechanical system design, and process improvement experience. Seeking Summer 2027 Engineering internship.

## Education

### Michigan Technological University

BS, Mechanical Engineering - Expected April 2028

GPA 3.85 | Dean's List | Pavlis Honors College

## Work Experience

### Coach - Engineering Fundamentals Fabrication Lab / MAE dept. / Fall 2026 – Present

- Taught 700+ incoming engineering to design and fabricate a manufacturable product.
- Provided hands-on guidance for safe use of equipment, including hand tools, 3D printers, power tools, and soldering irons.

### Undergraduate Research Assistant / MAE Dept. / Houghton, MI / Spring 2025 – Present

- Optimized a six-component Surface Plasmon Resonance stage using MATLAB, Arduino, and Breadboard circuits.
- Troubleshooted experimental hardware and refine test setups modeled in Onshape to improve reliability and repeatability.
- Established a reliable inventory of critical tools and identified safety hazards (OSHA 29 CFR 1910.1450).

### Heat Transfer Research Assistant / MAE Dept. / Lake Orion, MI / Summer 2025 & Summer 2026

- Use thermocouples to collect real world data and Arduino based data acquisition in excel to identify environmental trends.

### Admissions Associate / Merlin Entertainments / Auburn Hills, MI / Summer 2022

- Organized guests and large school fieldtrips, communicated and solved customer problems.

## Engineering Experience

### Nuclear Coolant Researcher - Alternative Energy Enterprise / Houghton, MI / Spring 2026 – Present

- Founding member- recruited initial team, established project for researching alternative coolant for fast reactors.
- Evaluated efficiency-improving design concepts (reduced corrosion), and support feasibility studies for engineering solutions.
- Collaborated with multidisciplinary team to evaluate technical concepts and develop clean-energy engineering solutions.

### Student Software Lead - FIRST Robotics Competition, Team 302 / Lake Orion / 2021-2024

- Mentored students in C++, contributing to 30% team growth and 90% overall retention of the 60+ member team.
- Programmed autonomous robots, tuned PID controllers for drive train and system actuators, robot system tester.

## Technical Projects

### Mobile Infant Incubator / Spring 2026 – Summer 2026

- Designed ASME-compliant orifice plate, performed Simcenter 3D FEA to verify deformation limits under operating conditions.
- Developed MATLAB code to quantify system uncertainty for ducting, heaters, and valves to identify dominant sources of error.
- Authored a Standard Operating Procedure for the heating test rig to document safe, repeatable testing and operation.

### CNC Lathe Machining / Fall 2025 – Spring 2026

- Designed a chess piece in CAD (Onshape) and created a manufacturing part drawing incorporating GD&T.
- Developed G-code (PathPilotHUB) to custom manufacture the part from 3" aluminum stock using a Tormach CNC lathe.

### Ferris Wheel Dynamic Response / Spring 2026 – Summer 2026

- Tuned PID controller through iterative testing to meet customer-defined stability, inertia, and acceleration requirements.
- Evaluated velocity and acceleration profiles against human tolerance limits to assess rider comfort and system performance.

### System Simulation with AMESim / Fall 2025 – Spring 2026

- Analyzed elevator system model using AMESim and compared simulation results with experimental response data in MATLAB.
- Calibrated accelerometer measurements to improve the accuracy of experimental system data and model validation.

## Technical Skills

- **Software:** MATLAB, MATLAB app designer, Onshape, Fusion 360, AutoCAD, Simcenter 3D, AMESim
- **Analysis & Testing:** FEA, uncertainty analysis, system simulation, experimental testing, process improvement
- **Manufacturing:** CNC machining (Tormach), GD&T, 3D printing (Bambu Lab), prototyping, mechanical assembly
- **Controls & Hardware:** PID control, Arduino, sensors, motors, electrical wiring, swerve modules

**Awards:** Dean's List, FIRST Robotics Autonomous Award, Robotics Rookie Allstar, Service-Learning Medal

**Leadership & Activities:** Engineering Ambassadors, Society of Women Engineers, American Society of Mechanical Engineers, National Advocacy Conference (Student Association for STEM Advocacy), Community Service (250+ logged hours)